

1. Find the equivalent force system at point A and the wrench location. Given $l = 6\text{ m}$, $f = 10 \frac{\text{kN}}{\text{m}}$, $\cos \phi = 0.8$, $x_B = 6\text{ m}$, $y_B = -5\text{ m}$ [10]

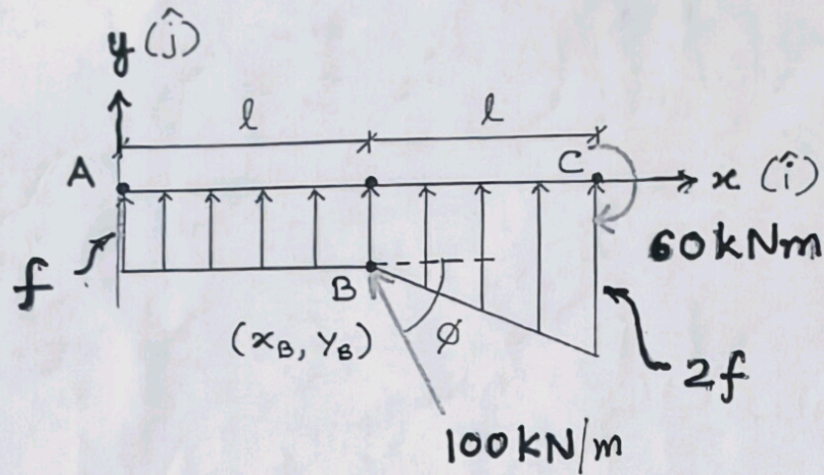


Fig. Q1

2. Two forces of the same magnitude P act on a cube of side a as shown. Replace the two forces by an equivalent wrench, and determine (a) the magnitude and direction of the resultant force $\underline{F}_R$ and the point where the axis of the wrench intersects the yz plane. [10]

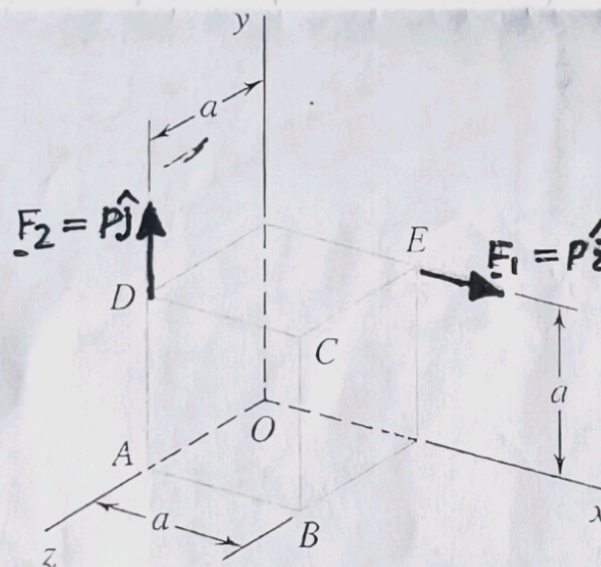


Fig. Q2

3. What are the conditions for principal-axes of the Inertia-tensor of a rigid body at Point A ($A \in RB$). Discuss about the existence of a set of 3 mutually perpendicular principal axes and importance thereof. [10]